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Precision Pays: A Fixed-Weight Boundary Loss Anchors the Precision End of a Recall–Precision Axis and Wins as the Consensus Veto under the Official BraTS-2023 Metrics

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Independent research

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Version 2.6 (2026-09-22) — measured provenance of the 55 exclusions, two statistics restated at full precision. No reported result changes.

Provenance. §3.1 read "1251 cases" while every statistic in this paper is computed on the nnU-Net study set of **1196**; the wording invited the reading that 1251 cases had been trained on. §3.1 now separates the download (1251 unified cases) from the study set (1196) and points at `analysis/DATASET_EXCLUSIONS.md`, which lists the 55 dropped cases one by one — trigger, git date of the triggering entry, and a re-measurement of their file integrity (55 excluded against 55 included controls: 0 unreadable file, 0 NaN/Inf, 0 out-of-set label, 0 divergent affine, (240, 240, 155) everywhere). The exclusion is a conservative, patient-level precaution recorded in March 2026, upstream of every training run and every metric; the French mirror ships as `analysis/DATASET_EXCLUSIONS_fr.md` with its CSV `analysis/exclusions_1251_to_1196.csv`.

Statistics restated (§4.4). The multi-seed nnU-Net validation Dice of the boundary loss is 0.9060 ± 0.0018 (per-seed 0.9062 / 0.9078 / 0.9042), previously printed as 0.9061 ± 0.0018 — the mean of per-seed values already rounded to 4 decimals — and its paired t-test against the baseline (0.9074 ± 0.0007) gives $t = -1.02$, $p = 0.42$, where the previously printed $t = 0.96$, $p = 0.44$ came from running that same test on values rounded first. Both readings sit far from significance ($df = 2$), so no conclusion changes; `data/nnunet_validation_dice_multiseed.json` now carries the six training-log measurements and both calculations.

Archive self-containment. Every path quoted in the manuscript resolves inside this archive: the trailing `References:` pointer named a path that does not exist here (`../references.bib` -> `references.bib`), and the three pinned-case figures now cite `data/patient_level_clean_BKD.json`, the artefact carrying the patient-level counts they quote.

Version 2.5 (2026-09-21) — author contributions. No number changed. An **Author contributions** section was added after §7, declaring the same roles as in Papers 1 and 3 (G.C.: study design, training, evaluations and analyses, writing; S.L.: research questions, methodological guidance, reviews of successive drafts). This banner is added now because the section went in without one — the public changelog did not describe the artefact.

Version 2.4 (2026-09-19) — title tightened, layout rules. No number changed. The hook is unchanged; the subtitle is shortened. The title block now starts a fresh page — it is never split across two pages — and the References are rendered at the end of the PDF (citations resolved from `references.bib`; no raw citation marker is left in the text). The section plan already matches the common skeleton shared with Papers 1 and 3 (Methods including Dataset and Evaluation, Results, Discussion, standalone Limitations, Conclusion).

Version 2.3 (2026-09-01) — authors and title. No number changed. By agreement between the two authors, Guillaume Cassez is first author and Stanislas Larnier second. The title now leads with what the loss buys — the precision end of the axis and the winning veto — the null result for the loss alone is stated inside the paper, not in its headline.

Version 2.2 (2026-08-22) — pre-review pass. No number changed. Author identifiers completed (Stanislas Larnier: HAL IdHAL `stanislas-larnier`; Guillaume Cassez: ORCID 0009-0007-0987-3931). The consensus results in the abstract and introduction now state their population: K∩B figures come from the

multiseed protocol (3 seeds × fold 0, pooled n = 720), the D∩K veto figure from the full 5-fold CV (n = 1196). Program-wide method confirmed on 2026-08-22: 3 seeds and 5 folds for every paper.

Version 2.1 (2026-08-11) — editorial pass. No number changed. The abstract now leads with the consensus result, which is the only significant finding of the study, and the null result for the boundary loss alone follows it; the trade the paper documents — locate the loss on the recall/precision axis, then spend it on agreement — is stated up front.

Version 2 (2026-06-14) — changelog. Supersedes v1. Evaluation redone under the **official BraTS-2023 metrics** at 300-epoch convergence on the full 5-fold CV (n = 1196). The boundary loss is **not significant vs the baseline** on any official metric; its distinctive trait is being the precision end of a recall/precision axis. New: we show Kervadec is an effective **veto** in the CC-consensus of the companion DistMap paper (its high specificity preserves recall as a veto). Added (2026-06-14): a **multiseed analysis** (3 seeds × fold 0, §4.4) that confirms the null-vs-baseline result, shows Kervadec **regularises** lesion-wise inter-seed variance (opposite of DistMap), and shows the K∩B consensus reproduces Paper 1's D∩B lesion-wise WT gain. Conclusions revised. (Concept DOI resolves to this version.)

Working draft — Paper 2 of the BRATS auxiliary-loss program. Quantitative claims come from `code/paper_stats.py` (`data/cv_paper_stats.json`); tables from `code/make_tables.py` (`table_paper2.{md,tex}`); figures from `code/make_figures.py`. Every official BraTS-2023 metric is reported; a pre-specified primary endpoint is separated from exploratory analyses; a non-significant / negative result is reported as such.

Abstract

Two models that must agree beat either of them alone. Under the official BraTS-2023 lesion-wise metrics, a parameter-free connected-component consensus — keep only the tumour components on which two models concur — removes about half of one model's false-positive lesions and produces the only significant improvements in this study: whole-tumour lesion-wise Dice **+3.89 pp** (p = 0.005) and HD95 **−15.66 mm** (p = 0.001) for the Kervadec∩baseline filter (3 seeds × fold 0, pooled n = 720), and HD95 **−9.95 mm** ($p = 2.3\times10^{-20}$; 5-fold CV, n = 1196) when the boundary-loss model vetoes the distance-map model of the companion Paper 1. The gain costs no training, no learned parameter and no hyper-parameter: it is the agreement that does the work, and it reproduces across both auxiliary losses (K∩B mirrors Paper 1's D∩B, +4.28 pp / −17.2 mm), which is what identifies the mechanism rather than the ingredient.

The ingredient itself is null, and we report that in full. The boundary loss of Kervadec et al. (Kervadec et al. 2019 ; Kervadec et al. 2021) is designed to reduce contour error in highly unbalanced segmentation. We test whether adding it (fixed weight $\lambda = 0.05$, acting directly on the logits, no auxiliary head) to a strong MedNeXt/nnU-Net backbone improves adult glioma segmentation, evaluated under the **complete** official BraTS-2023 metric set over a 5-fold held-out cross-validation (n = 1196). Against the baseline, the boundary loss is **not significant on the pre-specified primary endpoints** (lesion-wise Dice $\Delta = +0.000$, Holm p = 0.34; lesion-wise HD95 $\Delta = -0.58$ mm, Holm p = 0.31), **Dice-neutral** on legacy overlap (avg $\Delta = +0.000$, p = 0.06), **null on legacy region HD95** (avg $\Delta = -0.09$ mm, p = 0.11), and **detection-neutral** (region sensitivity p = 0.26, specificity p = 0.08). In short, a fixed-weight boundary penalty is statistically indistinguishable from the baseline on a backbone that already has deep supervision — a negative result reported here in full, with a weight-selection robustness check that rules out mistuning.

What the loss is good for is being a veto. Its one distinctive, robust standalone property appears **relative to a signed-distance (SDT) auxiliary head**: the boundary loss is significantly **more specific** (region specificity r = +0.27, $p = 2.6\times10^{-16}$) and produces **fewer spurious tumour-core/enhancing lesions** (FP lesions TC $\Delta = -0.066$, ET $\Delta = -0.195$, both Holm p = 0.004), at a sensitivity cost — the mirror image of the SDT head's recall bias. That precision is exactly what makes it an effective corroborator: as a veto it removes false-positive lesions while barely touching recall (0.006 added missed lesions per case against 0.009 for a baseline veto). An independent **multiseed** analysis (3 seeds × fold 0) confirms the null-vs-baseline result (t-test p = 0.42) and shows the boundary loss **regularises** lesion-wise inter-seed variance (σ LW Dice 3–12× lower than baseline by region) — the opposite of DistMap. The practical reading: locate an auxiliary loss on the recall–precision axis, then spend it on agreement rather than on the leaderboard.

1. Introduction

Requiring two segmentation models to *agree* can be worth more than either model on its own. Under the official BraTS-2023 lesion-wise metrics, a connected-component consensus — keep only the tumor components on which two models concur — removes about half of one model's false-positive lesions and yields a significant whole-tumor gain: lesion-wise Dice **+3.89 pp** ($p = 0.005$) and HD95 **−15.66 mm** ($p = 0.001$) for the Kervadec∩baseline filter (3 seeds × fold 0, pooled $n = 720$), and **−9.95 mm** HD95 ($p = 2.3\times10^{-20}$; 5-fold CV, $n = 1196$) when the boundary-loss model instead vetoes the distance-map model of the companion Paper 1. We lead with this because it is the finding that moves the clinically relevant, lesion-level metric.

The improvement comes from the agreement, not from any single loss. The lesion-wise metric penalises each spurious component individually, so discarding components that a second model does not support is precisely the operation it rewards. The boundary loss contributes here through one specific, robust property — high specificity — which makes it an effective **veto**: it removes false-positive lesions while barely touching recall (0.006 added missed lesions per case versus 0.009 for a baseline veto). The gain is carried by the consensus rule and holds regardless of which auxiliary loss is the primary model — K∩B mirrors Paper 1's D∩B (+4.28 pp / −17.2 mm) — which indicates the mechanism, rather than the loss term, is doing the work.

The honest twist is that **neither model, taken on its own, significantly improves anything.** The boundary loss (Kervadec et al. 2019 ; Kervadec et al. 2021) — a level-set contour penalty that has helped weak baselines — adds nothing measurable when bolted onto a **modern, strong** 3D backbone (MedNeXt (Roy et al. 2023) within nnU-Net (Isensee et al. 2021)) that already has deep supervision and a Dice + cross-entropy objective: against the baseline it is not significant on any pre-specified primary or region-averaged official metric (lesion-wise Dice $\Delta = +0.000$, Holm $p = 0.34$; lesion-wise HD95 $\Delta = -0.58$ mm, $p = 0.31$), and a multiseed check (3 seeds × fold 0) confirms the null (t-test $p = 0.42$). The value is not in the ingredient but in the agreement.

We therefore report, honestly and in full, both sides: a transparent negative result for the fixed-weight boundary loss on its own — guarded against the mistuning objection by a weight-selection robustness check — and the mechanism-level finding that its precision makes it a useful consensus component. Contributions: 1. A convergence-time, 5-fold evaluation ($n = 1196$) of a fixed-weight boundary loss on a strong 3D glioma backbone under the complete official BraTS-2023 metric set — a transparently reported **negative result** over the baseline, with a robustness check ruling out weight mistuning. 2. The one robust standalone signal — that the boundary loss is the **precision-oriented** counterpart of an SDT auxiliary head (significantly more specific, fewer spurious TC/ET lesions, lower sensitivity) — locating both auxiliary losses on a single recall–precision axis. 3. The consensus finding: Kervadec's specificity makes it an effective veto (D∩K) and a viable primary model (K∩B) in a connected-component consensus that significantly improves the lesion-wise whole-tumor metrics, with the gain attributable to the agreement mechanism rather than the loss.

2. Related work

Boundary-integral losses (Kervadec et al. 2019 ; Kervadec et al. 2021) vs distance-map regression (Ma et al. 2020 ; Xue et al. 2020 ; Dangi et al. 2019) are two families of shape/boundary-aware objectives; the companion Paper 1 studies the latter. Backbone and training: nnU-Net (Isensee et al. 2021), MedNeXt (Roy et al. 2023), Dice/GDL (Milletari et al. 2016 ; Sudre et al. 2017). Data and metrics: BraTS (Menze et al. 2015 ; Bakas et al. 2017 ; Baid et al. 2021); lesion-wise metrics (Moawad et al. 2023 ; Saluja et al. 2023); surface distance (Nikolov et al. 2021).

3. Methods

3.1 Dataset and backbone

Identical to the companion Paper 1. **Data.** BraTS-2023 adult glioma (GLI) (Baid et al. 2021): **1251 unified cases downloaded, 1196 in the study set** — the study set is what every training run and every metric below uses. The 55 dropped cases are listed one by one (trigger, git date of the triggering entry, re-measurement of their file integrity: 55 excluded vs 55 included controls, 0 unreadable file, 0 NaN/Inf, 0 out-of-set label, 0 divergent affine) in `analysis/DATASET_EXCLUSIONS.md`. The drop is a patient-level precaution recorded in March 2026, upstream of any training run and any metric; no published figure changes. **Backbone.** Official nnU-Net 5-fold splits, MedNeXt-B

via nnU-Net v2 (`3d_fullres`, `nnUNetPlans_96GB_mednext`), Dice + cross-entropy with deep supervision, 300 epochs/fold, RTX PRO 6000. All methods share folds (paired statistics, n = 1196).

3.2 Boundary loss

We add the Kervadec boundary term as `seg_loss + λ · mean(softmax(logits) · SDT_signed)` (θ_0 = 30, fixed λ = 0.05), acting directly on the segmentation logits with **no auxiliary head**. Implementation: `code/nnUNetTrainerMedNeXtKervadec.py`; signed SDT cache precomputed for all cases.

3.3 Weight selection and robustness

$\lambda \in \{0.05, 0.15, 0.5, 1.5, 5.0\}$ was screened on a short schedule (50 epochs, fold 0); λ = 0.05 maximised EMA pseudo-Dice. Because that criterion is overlap-based, we additionally re-rank the screening on sentinel-capped HD95 (median + mean): λ = 0.05 is already near HD95-optimal (lowest whole-tumor median HD95, fewest catastrophic misses), and larger λ **degrades** HD95 (median rises monotonically; λ = 5.0 collapses). The null region-wise result below is therefore not a weight-selection artefact.

3.4 Evaluation

Identical official-metric protocol and statistics as Paper 1 (official BraTS-2023-Metrics, GLI parameters; paired two-sided Wilcoxon; rank-biserial r with + favoring Kervadec and its bootstrap CI; Holm correction across regions; Benjamini–Hochberg FDR across the whole 216-test exploratory family; pre-specified primary endpoint = the two region-averaged official ranking metrics). We evaluate both Kervadec vs baseline and Kervadec vs the SDT head (DistMap). Pipeline: `code/eval_lesionwise_kervadec.py` → `code/paper_stats.py`.

4. Results

4.1 Versus baseline — no significant effect anywhere central

- **Primary endpoints (null):** lesion-wise Dice Δ = +0.000 (Holm p = 0.34); lesion-wise HD95 Δ = −0.58 mm (r = +0.06, raw p = 0.155, Holm p = 0.31).
- **Overlap (Dice-neutral):** legacy Dice avg Δ = +0.000 (p = 0.063; Table 1). A single nominal effect — tumor-core legacy Dice Δ = +0.001 (p = 0.005, Holm 0.015†) — is statistically significant but negligible in magnitude.
- **Boundary (null):** legacy region HD95 avg Δ = −0.09 mm (r = +0.07, p = 0.109).
- **Detection (neutral):** region sensitivity avg Δ = −0.001 (p = 0.256); specificity avg Δ = +0.000 (r = +0.06, p = 0.079). Lesion FP/FN counts vs baseline are all non-significant (Table 2). The boundary loss does not separate from the baseline.

Pre-specified primary endpoints (Kervadec vs Baseline, region-averaged official ranking metrics, Holm-corrected across the 2-metric family):

- Lesion-wise Dice: Δ = +0.000, r = +0.032, raw p = 0.339, Holm p = 0.339 → not significant
- Lesion-wise HD95: Δ = -0.582 mm, r = +0.056, raw p = 0.155, Holm p = 0.311 → not significant

Table 1 — Official ranking & legacy metrics (Kervadec vs Baseline)

Δ = Kervadec − Baseline. r = matched-pairs rank-biserial (+ favours Kervadec). * raw p<0.05; † Holm-corrected p<0.05 (across the 3 regions). n=1196.

Metric	Region	Kervadec	Baseline	Δ	r	win/loss	p	p(Holm)
Lesion-wise Dice	WT	0.811	0.812	-0.001	+0.03	614/580	0.352	0.658
	TC	0.869	0.870	-0.001	+0.07	643/543	0.039*	0.118
	ET	0.800	0.798	+0.003	+0.03	603/557	0.329	0.658
	avg	0.827	0.826	+0.000	+0.03	635/559	0.339	–
Lesion-wise HD95 (mm)	WT	53.87	53.09	+0.78	+0.06	337/289	0.216	0.631
	TC	24.33	24.68	-0.35	-0.01	222/225	0.889	0.889
	ET	42.71	44.89	-2.18	+0.07	200/173	0.210	0.631
	avg	40.31	40.89	-0.58	+0.06	458/388	0.155	–
Legacy Dice	WT	0.936	0.935	+0.000	+0.03	619/575	0.397	0.648
	TC	0.918	0.918	+0.001	+0.09	645/542	0.005*	0.015†
	ET	0.870	0.869	+0.000	+0.03	606/557	0.324	0.648
	avg	0.908	0.907	+0.000	+0.06	646/548	0.063	–
Legacy HD95 (mm)	WT	5.77	5.79	-0.02	+0.06	291/269	0.252	0.589
	TC	6.02	6.00	+0.01	+0.06	214/199	0.282	0.589
	ET	11.36	11.62	-0.26	+0.08	172/145	0.196	0.589
	avg	7.72	7.81	-0.09	+0.07	420/354	0.109	–

Table 2 — Lesion detection (Kervadec vs Baseline)

Δ = Kervadec − Baseline. r = matched-pairs rank-biserial (+ favours Kervadec). * raw p<0.05; † Holm-corrected p<0.05 (across the 3 regions). n=1196.

Metric	Region	Kervadec	Baseline	Δ	r	win/loss	p	p(Holm)
FP lesions	WT	0.334	0.337	-0.003	+0.00	112/112	0.945	1.000
	TC	0.122	0.132	-0.010	-0.02	38/46	0.725	1.000
	ET	0.641	0.718	-0.077	+0.14	80/74	0.158	0.474
	avg	0.366	0.396	-0.030	+0.02	172/176	0.777	–
FN lesions	WT	0.079	0.080	-0.001	+0.11	5/4	0.739	0.739
	TC	0.033	0.037	-0.003	+0.43	5/2	0.206	0.412
	ET	0.038	0.042	-0.004	+0.56	6/2	0.132	0.395
	avg	0.050	0.053	-0.003	+0.37	10/7	0.244	–
Sensitivity	WT	0.929	0.929	-0.000	-0.03	574/618	0.438	0.438
	TC	0.919	0.921	-0.002	+0.04	636/544	0.193	0.386
	ET	0.876	0.877	-0.000	+0.07	623/532	0.051	0.154
	avg	0.908	0.909	-0.001	+0.04	627/567	0.256	–
Specificity	WT	1.000	1.000	+0.000	+0.07	626/566	0.039*	0.118
	TC	1.000	1.000	+0.000	+0.00	573/607	0.908	0.908
	ET	1.000	1.000	-0.000	-0.04	584/593	0.259	0.517
	avg	1.000	1.000	+0.000	+0.06	616/578	0.079	–

4.2 Versus the SDT head — a precision-oriented mirror (exploratory, robust)

The boundary loss differs sharply from the SDT auxiliary head (Table 3–4, Fig. 1–2): - **More specific:** region specificity r = +0.27 (avg $\mathbf{p = 2.6\times10^{-16}}$; all regions Holm†). - **Fewer spurious lesions:** FP lesions TC Δ = −0.066 (p = 0.002, Holm 0.004†), ET Δ = −0.195 (p = 0.001, Holm 0.004†). - **Lower sensitivity / more missed lesions:** region sensitivity r = −0.18 (avg p = 1.1×10^{-7}). - **Better enhancing-tumor lesion-wise HD95:** Δ = −4.05 mm (p = 0.015, Holm 0.046†) and lesion-wise ET Dice Δ = +0.008 (p = 0.020).

The boundary loss thus occupies the **precision** end of a recall–precision axis whose recall end is the SDT head (Paper 1); neither beats the baseline on the official ranking metrics.

Pre-specified primary endpoints (Kervadec vs DistMap, region-averaged official ranking metrics, Holm-corrected across the 2-metric family):

- Lesion-wise Dice: Δ = +0.003, r = +0.033, raw p = 0.324, Holm p = 0.647 → not significant
- Lesion-wise HD95: Δ = -1.700 mm, r = +0.027, raw p = 0.487, Holm p = 0.647 → not significant

Table 3 — Official ranking & legacy metrics (Kervadec vs DistMap)

Δ = Kervadec − DistMap. r = matched-pairs rank-biserial (+ favours Kervadec). * raw p<0.05; † Holm-corrected p<0.05 (across the 3 regions). n=1196.

Metric	Region	Kervadec	DistMap	Δ	r	win/loss	p	p(Holm)
Lesion-wise Dice	WT	0.811	0.816	-0.005	-0.01	606/589	0.785	0.785
	TC	0.869	0.862	+0.008	+0.05	613/576	0.161	0.323
	ET	0.800	0.792	+0.008	+0.08	618/545	0.020*	0.061
	avg	0.827	0.823	+0.003	+0.03	616/579	0.324	–
Lesion-wise HD95 (mm)	WT	53.87	51.11	+2.76	-0.03	328/332	0.486	0.971
	TC	24.33	28.14	-3.81	+0.03	225/232	0.626	0.971
	ET	42.71	46.76	-4.05	+0.14	231/193	0.015*	0.046†
	avg	40.31	42.01	-1.70	+0.03	450/418	0.487	–
Legacy Dice	WT	0.936	0.936	-0.000	+0.01	607/588	0.786	0.786
	TC	0.918	0.917	+0.001	+0.05	617/572	0.111	0.332
	ET	0.870	0.871	-0.001	+0.05	612/553	0.158	0.332
	avg	0.908	0.908	-0.000	+0.03	616/579	0.362	–
Legacy HD95 (mm)	WT	5.77	5.51	+0.26	-0.01	284/292	0.838	1.000
	TC	6.02	5.89	+0.13	-0.05	206/222	0.384	1.000
	ET	11.36	11.54	-0.18	+0.01	183/174	0.873	1.000
	avg	7.72	7.65	+0.07	-0.02	400/393	0.646	–

Table 4 — Lesion detection (Kervadec vs DistMap)

Δ = Kervadec − DistMap. r = matched-pairs rank-biserial (+ favours Kervadec). * raw p<0.05; † Holm-corrected p<0.05 (across the 3 regions). n=1196.

Metric	Region	Kervadec	DistMap	Δ	r	win/loss	p	p(Holm)
FP lesions	WT	0.334	0.297	+0.037	-0.10	116/132	0.131	0.131
	TC	0.122	0.188	-0.066	+0.39	61/34	0.002*	0.004†
	ET	0.641	0.836	-0.195	+0.26	110/73	0.001*	0.004†
	avg	0.366	0.440	-0.075	+0.08	206/185	0.123	–
FN lesions	WT	0.079	0.075	+0.004	-0.34	4/9	0.166	0.472
	TC	0.033	0.030	+0.003	-0.50	2/6	0.157	0.472
	ET	0.038	0.035	+0.003	-0.03	4/7	0.366	0.472
	avg	0.050	0.047	+0.003	-0.30	10/16	0.064	–
Sensitivity	WT	0.929	0.931	-0.002	-0.20	486/706	2.4e-09*	7.3e-09†
	TC	0.919	0.921	-0.002	-0.08	551/631	0.018*	0.018†
	ET	0.876	0.880	-0.004	-0.15	494/659	1.6e-05*	3.2e-05†
	avg	0.908	0.910	-0.002	-0.18	495/700	1.1e-07*	–
Specificity	WT	1.000	1.000	+0.000	+0.25	705/488	1.2e-13*	3.5e-13†
	TC	1.000	1.000	+0.000	+0.17	654/529	6.6e-07*	6.6e-07†
	ET	1.000	1.000	+0.000	+0.23	684/498	3.2e-12*	6.5e-12†
	avg	1.000	1.000	+0.000	+0.27	732/463	2.6e-16*	–

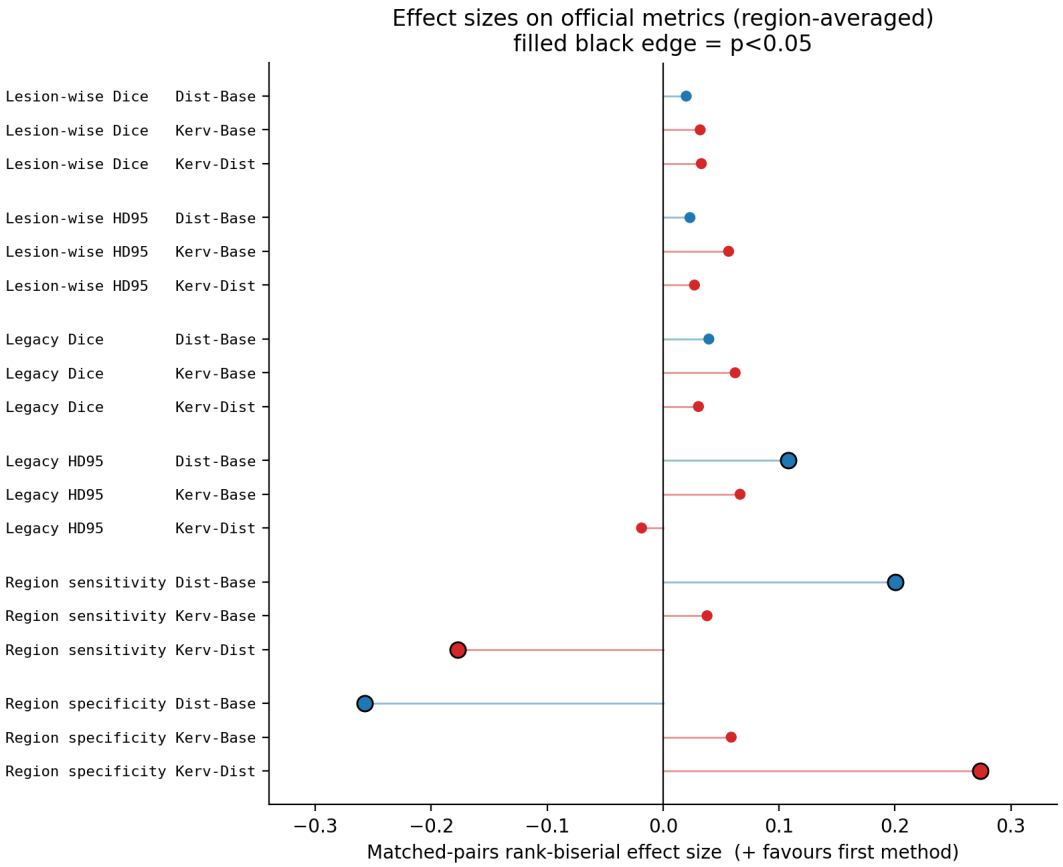


Figure 1: Figure 1 — Rank-biserial effect sizes (region-averaged) for each comparison, with confidence intervals: the boundary loss stays inside the noise against the baseline on the primary endpoints, and separates from the SDT head on specificity and spurious lesions.

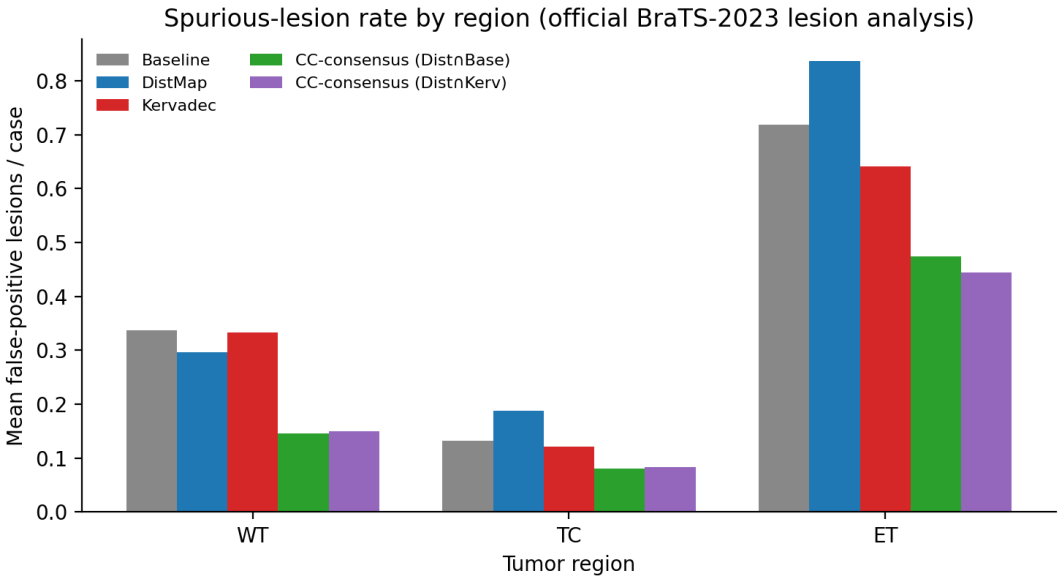


Figure 2: Figure 2 — Spurious-lesion rate by region × method: the precision-bias signature of the boundary loss (fewer spurious TC/ET lesions than the SDT head, at a recall cost).

Three pinned patients of the companion 3D viewer (https://guillaume-cassez.fr/imagerie-medicale/brats/2023-distance-map/viewer/, official regime: white background, BraTS-2023 component cleaning, viewer region colors) put faces on these patient-level counts: a case the boundary loss rescues when both other models fail (Figure 3), a case it fails and the baseline dominates (Figure 4), and a case the SDT head rescues (Figure 5).

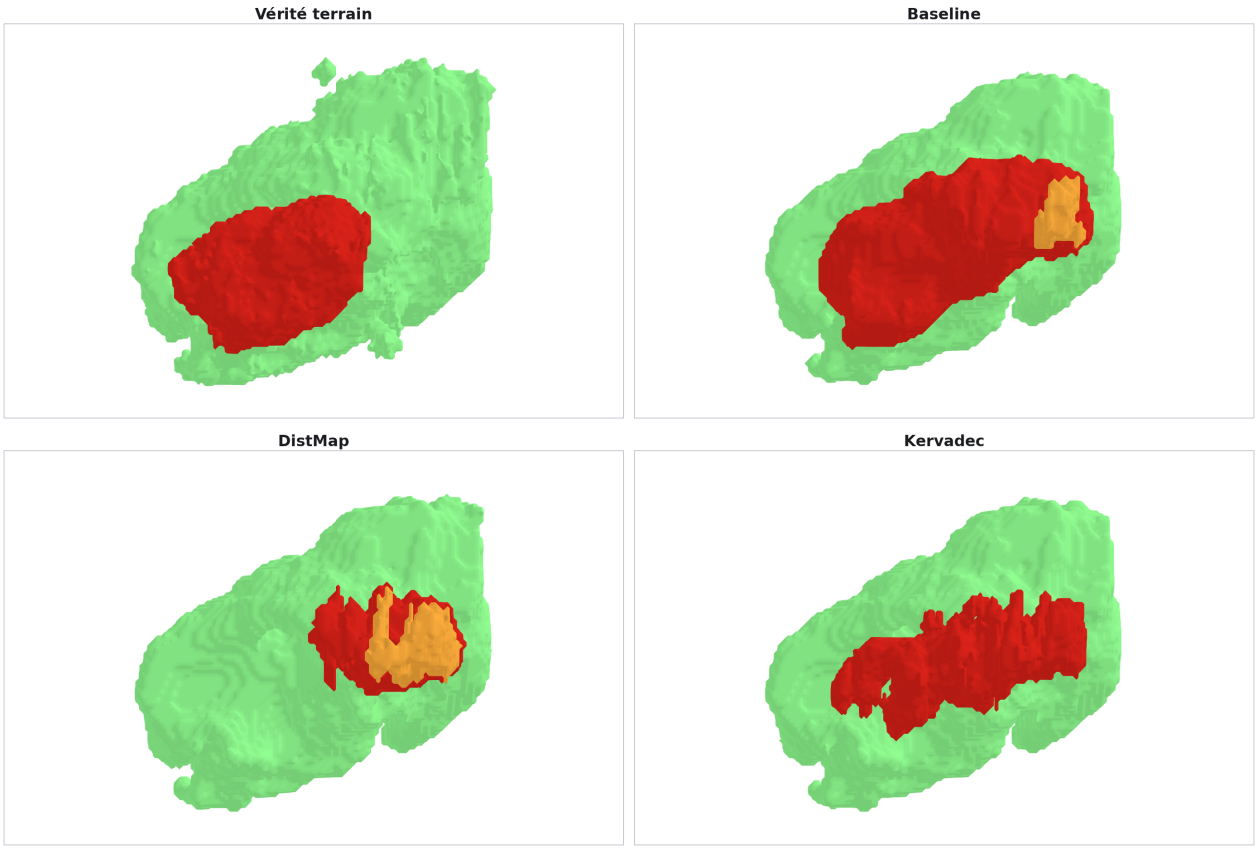


Figure 3: Figure 3 — Pinned case P2A (`BraTS-GLI-01486-000`, fold 1): **Kervadec rescues.** Kervadec (0.822) ≫ Baseline (0.526) > DistMap (0.324): the boundary loss captures the tumor where both others fail — 415/1196 patients (34.7 %) see Kervadec beat both Baseline and DistMap on lesion-wise Dice (official clean regime, mean over the 3 regions). Patient-level counts and pinned-case scores: `data/patient_level_clean_BKD.json`. Exact rendering of the companion 3D viewer: white background, official filter, sagittal view, brain masked.

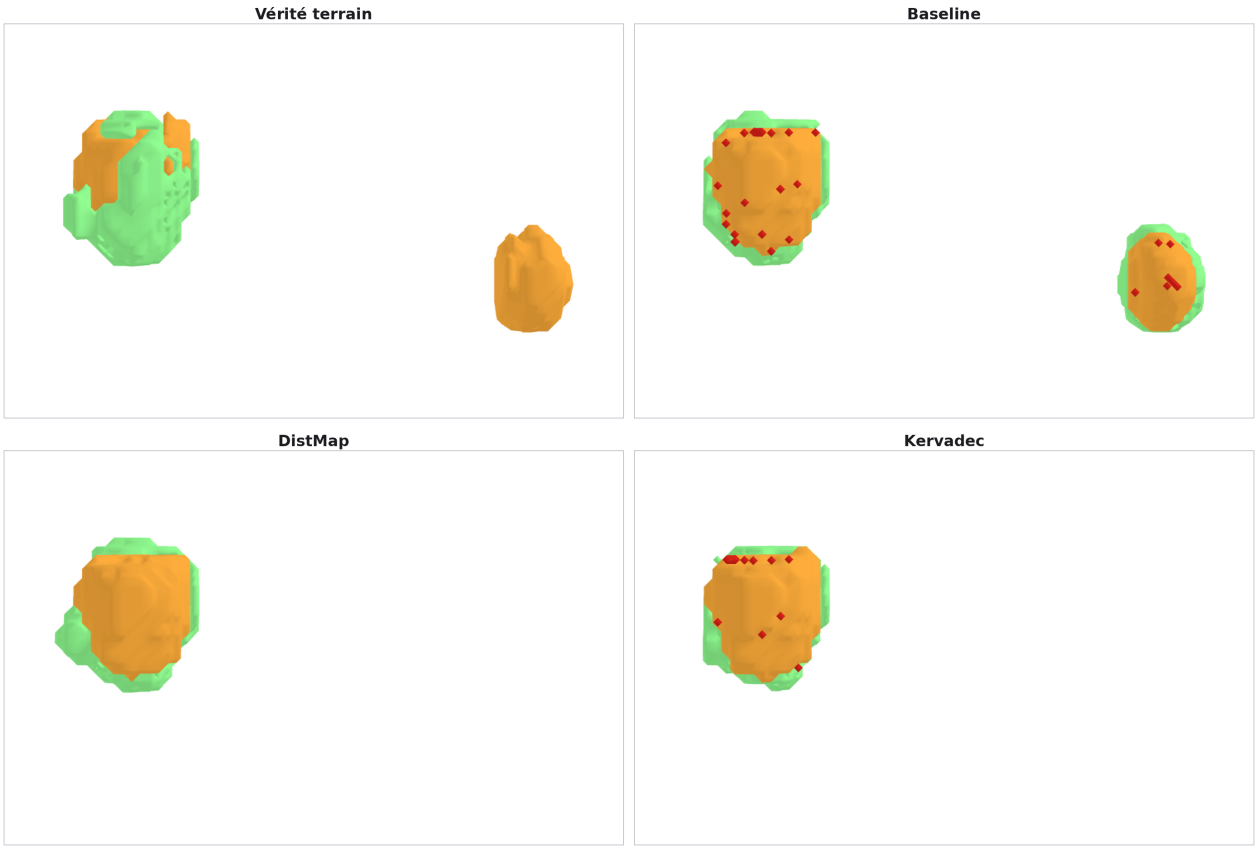


Figure 4: Figure 4 — Pinned case P2B (`BraTS-GLI-00732-001`, fold 4): **Kervadec fails.** Baseline (0.897) $\gg$ Kervadec (0.476) $\approx$ DistMap (0.473) — the mirror of the previous case: 346 patients (28.9 %) see Baseline beat both auxiliary losses. Too strict on contours, the boundary loss misses here tumor portions that regional CE + Dice captures; on this patient, no auxiliary loss helps. Patient-level counts and pinned-case scores: `data/patient_level_clean_BKD.json`.

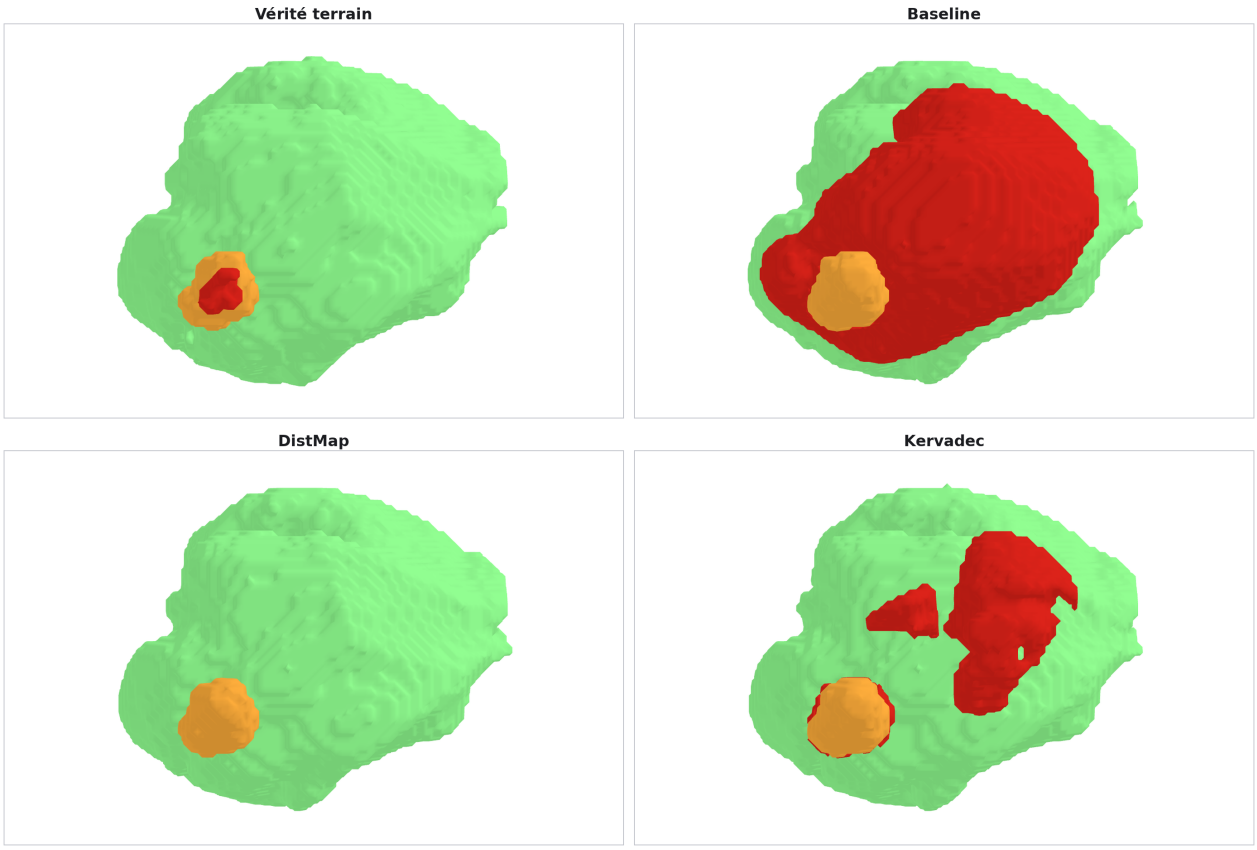


Figure 5: Figure 5 — Pinned case P2C (`BraTS-GLI-00525-000`, fold 4): **DistMap rescues.** DistMap (0.905) > Kervadec (0.698) > Baseline (0.623) — 433 patients (36.2 %) see DistMap beat both other experts: the signed distance-map regression recalls structures that the regional loss smooths away. Patient-level counts and pinned-case scores: `data/patient_level_clean_BKD.json`.

4.3 Kervadec as a consensus veto

Its high specificity makes Kervadec an effective veto in the CC-consensus studied in the companion DistMap paper. Filtering DistMap's components by Kervadec agreement (`CC(D∩K)`) removes ~half of DistMap's false-positive lesions (0.44 $\rightarrow$ 0.23/case) while preserving recall marginally better than a baseline veto (added missed lesions 0.006 vs 0.009/case), and yields a configuration that significantly beats the baseline on the official ranking metrics (lesion-wise HD95 −9.95 mm, p = 2.3×10^{-20}). The boundary loss thus contributes not as a standalone improvement but as a precision-oriented veto component — its high specificity is exactly the property a veto needs.

4.4 Seed robustness (3 seeds, fold 0)

The CV evaluation (§4.1–4.3) uses seed 42 across all 5 folds. To separate a robust signal from an initialisation fluctuation and to measure the boundary loss's inter-seed variance, three independent trainings (seeds 1, 2, 3) were run for baseline and Kervadec (λ = 0.05) on fold 0 (n $\approx$ 240), 300 epochs each — mirroring the companion DistMap multiseed protocol (Paper 1, §4.6). Source: `code/evaluate_multiseed_kervadec.py` (`data/multiseed_stats.json`).

- **Null vs baseline, again.** nnU-Net validation Dice: baseline 0.9074 ± 0.0007 vs Kervadec 0.9060 ± 0.0018 (per-seed 0.9062 / 0.9078 / 0.9042); paired t-test t = −1.02, p = 0.42 at full precision — artefact `data/nnunet_validation_dice_multiseed.json`, which reads the six training logs and carries both calculations (the rounded-input one, quoted before v2.6, and this one). Under the official metrics, Kervadec and

baseline stay indistinguishable on legacy Dice (all p > 0.40 Wilcoxon). Two sub-significant trends favour better boundary placement: LW HD95 WT −5.35 mm (p = 0.058) and TC −4.39 mm (p = 0.099).

- **Kervadec regularises lesion-wise variance.** On overlap Dice the three configs are comparably stable; but on the official lesion-wise metrics Kervadec's inter-seed σ is markedly **lower** than baseline — σ(LW Dice) 0.0080→0.0021 (WT, ×3.9), 0.0067→0.0024 (TC, ×2.8), 0.0109→0.0009 (ET, ×12). This is the **opposite** of DistMap, which was more unstable than baseline on the same protocol (Paper 1, §4.6). Suppressing spurious components tightens the lesion-wise spread — the signature of the loss's precision bias (§4.2).
- **K∩B consensus reproduces the D∩B gain.** Keeping Kervadec connected components overlapping a baseline prediction (K∩B) significantly improves LW Dice WT (+3.89 pp, p = 0.005 Wilcoxon) and LW HD95 WT (−15.66 mm, p = 0.001), at no legacy-Dice cost — close to the D∩B gain on the same protocol (Paper 1, §4.6: +4.28 pp / −17.2 mm). The connected-component consensus rule produces the lesion-wise WT improvement **regardless of which auxiliary loss is the primary model**: the gain is carried by the consensus mechanism, not the loss. This complements §4.3 — Kervadec is useful in consensus both as a **veto** (D∩K, filtering DistMap) and as a **primary model** (K∩B, filtered by baseline).

5. Discussion

A fixed-weight boundary loss does not help a strong backbone. With deep supervision and a combined Dice + CE objective already present, an added contour penalty leaves overlap, region boundary, and detection unchanged relative to the baseline. This is a useful negative result that bounds the expected benefit of bolt-on boundary penalties for state-of-the-art 3D glioma backbones — reported transparently, with a weight-selection robustness check that excludes mistuning.

Its distinctive trait is precision, visible only by contrast. Against the SDT head, the boundary loss is significantly more specific and produces fewer spurious tumor-core/enhancing lesions, at a sensitivity cost. Read together with Paper 1, the two auxiliary losses are mirror images on a recall–precision axis: the SDT head trades specificity for recall, the boundary loss trades recall for specificity, and neither improves the official ranking metrics over the baseline. This contrast — not a win over the baseline — is the contribution.

6. Limitations

The main comparison (vs baseline) is null; we present it as such. The specificity/FP advantages hold only against the SDT head and are exploratory (primary endpoints null), though robust ($p \leq 10^{-7}$, surviving Holm). Region sensitivity/specificity are voxel-level. Fold-0 baseline/DistMap predictions come from a headline run on external media. Only one fixed weight was carried to full CV, justified by the screening robustness check.

7. Conclusion

On a strong MedNeXt/nnU-Net glioma backbone, a fixed-weight Kervadec boundary loss yields **no significant improvement over the baseline** under any official BraTS-2023 metric. Its only robust, distinctive property is being the **precision-oriented mirror** of a signed-distance auxiliary head — more specific, fewer spurious lesions, lower recall — a property that also makes it a *regulariser* of lesion-wise variance and an effective consensus *veto*. The measurable gain comes not from the loss but from the connected-component consensus filter, which yields the same lesion-wise WT improvement whether the loss serves as a veto (D∩K) or as the primary model (K∩B). Reported in full, this delimits when boundary losses help (not here, over a strong baseline) and what they actually change (the recall–precision operating point, and training stability).

Author contributions

Guillaume Cassez (lead author): study design, model training, evaluations and analyses, manuscript writing. **Stanislas Larnier**: formulation of the research questions, methodological guidance, careful reviews of successive drafts of the paper. Both authors approved the final version and the author order.

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